AutoVSR: Automatic Visual-to-Symbolic Reasoning for Symbolic Expression Generation from Circuit Schematic

Zhe Xiao*, Longfei Li*, Xu He#, Haoying Wu, Zixing Zhang, Mingyu Liu

Hunan University | Wuhan University of Technology | Huazhong University of Science and Technology

Paper/AutoVSR Code/AutoVSR

Verified executable IR + tool-augmented symbolic reasoning for reliable circuit analysis

Background

Symbolic expressions (e.g., transfer functions) are fundamental for analyzing and predicting circuit behavior.

• Symbolic expressions are crucial for characterizing and predicting circuit behavior.

• Deriving expressions directly from circuit schematics is challenging because it requires both visual understanding and correct multi-step symbolic reasoning.

• Existing end-to-end VLM approaches lack explicit intermediate verification, while symbolic toolchains such as SymPy and Lcapy require structured symbolic inputs rather than raw schematic images.

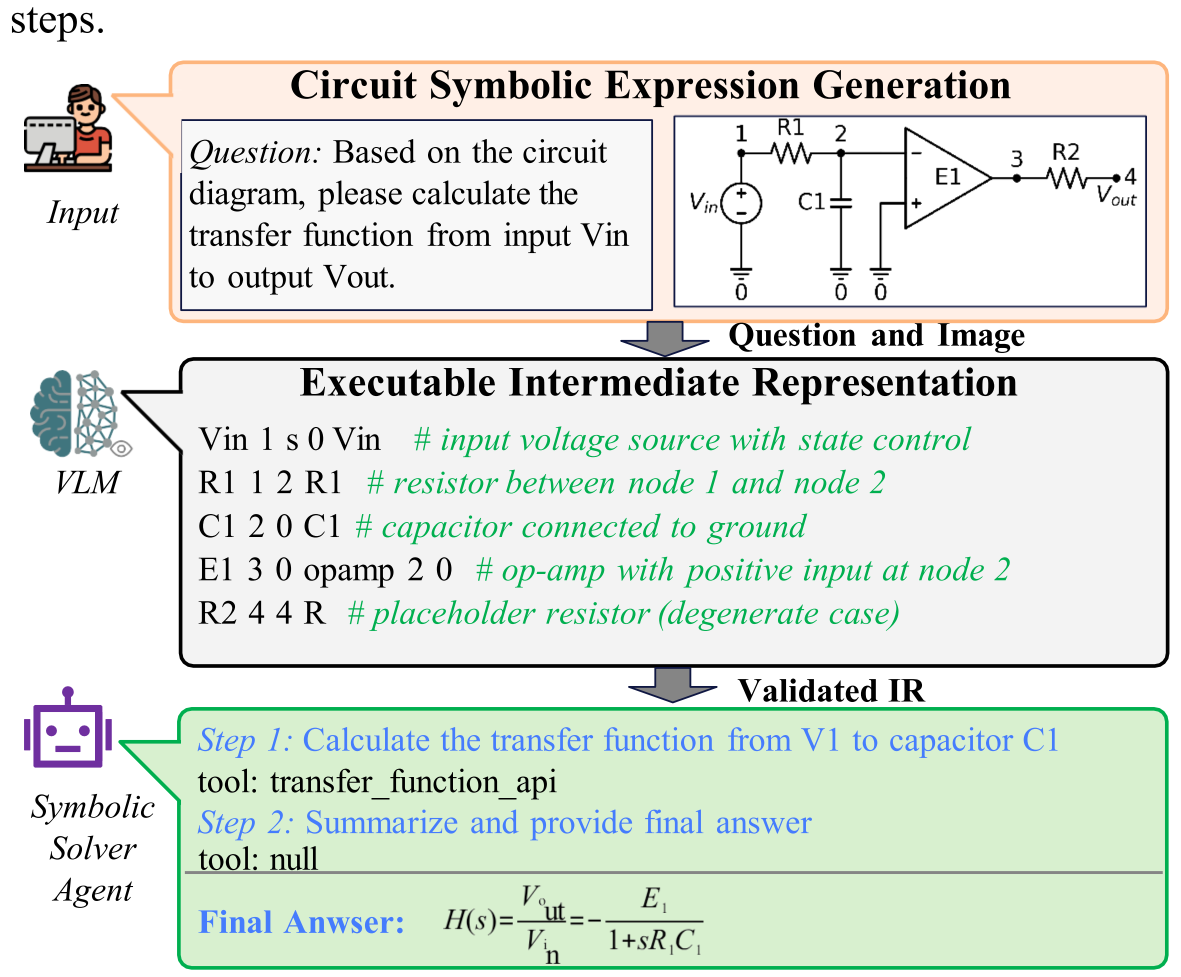
Challenges & Motivation

Challenge 1: Reliable visual-to-symbolic construction

Circuit schematics do not directly provide machine-readable device types, connectivity, or variable definitions; errors in reconstruction propagate to later reasoning.

Challenge 2: Correct multi-step symbolic derivation

Symbolic derivation is vulnerable to errors in sign conventions, equation formulation, and algebraic manipulation. Such errors can propagate and accumulate across multiple reasoning steps.



AutoVSR bridges schematic images and symbolic expressions through an executable, verifiable intermediate representation and a tool-based symbolic solver.

Contributions

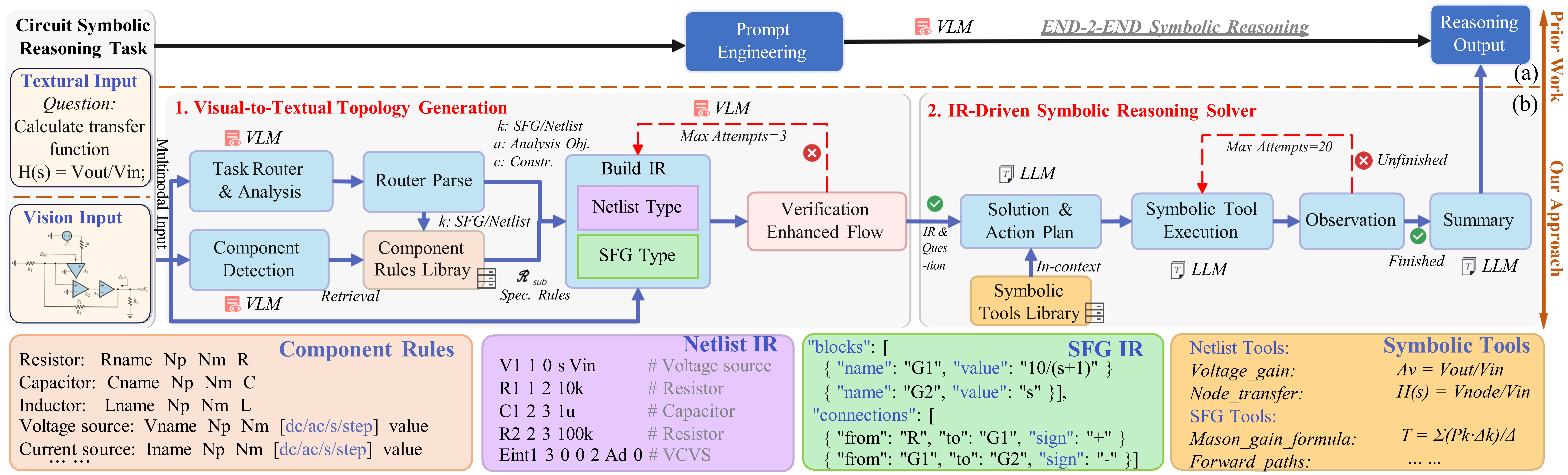
◆ Propose AutoVSR, a visual-to-symbolic reasoning framework forsymbolic expression generation from circuit schematics.

◆Introduce an Executable IR with two forms: Netlist IR for schematic circuits and SFG IR for system-level signal-flow graphs.

◆Design a verification-enhanced visual-to-textual topology generation stage with task routing, component rule retrieval, and iterative error-correction feedback.

◆ Build an IR-driven symbolic reasoning solver with plan-and-execute reasoning and a symbolic tool library, achieving strong gains inaccuracy, efficiency, and interpretability.

Overview



Experiments

Table 1. Performance Comparison of AutoVSR and End-to-End VLMs on Transfer Function Generation Across Different Task Types.

Model Type	Method	Type 1 Acc. (%)	Type 1 Imp. ↑	Type 2 Acc. (%)	Type 2 Imp. ↑	Type 3 Acc. (%)	Type 3 Imp. ↑	Type 4 Acc. (%)	Type 4 Imp. ↑	Type 5 Acc. (%)	Type 5 Imp. ↑	Overall Acc. (%)	Overall Imp. ↑
Closed Source	Gemini-3-Flash	37.95	+46.99	35.67	+53.51	23.06	+51.29	11.93	+72.73	3.95	+83.77	23.40	+59.45
	AutoVSR(Gemini-3-Flash)	**84.94**		**89.18**		**74.35**		**84.66**		**87.72**		**82.85**	
	GPT-5-mini	9.04	+83.13	9.36	+68.13	7.97	+23.50	13.64	+40.34	16.23	+14.91	10.54	+42.51
	AutoVSR(GPT-5-mini)	**92.17**		**77.49**		**31.47**		**53.98**		**31.14**		**53.05**	
	Qwen3-VL-Plus	18.07	+74.70	19.30	+74.85	13.36	+35.78	19.89	+50.00	15.79	+32.02	16.64	+51.38
	AutoVSR(Qwen3-VL-Plus)	**92.77**		**94.15**		**49.14**		**69.89**		**47.81**		**68.02**	
	Claude-Haiku-4.5	9.64	+64.46	10.53	+68.71	10.56	+18.10	9.09	+15.91	5.70	+14.04	9.67	+35.10
	AutoVSR(Claude-Haiku-4.5)	**74.10**		**79.24**		**28.66**		**25.00**		**19.74**		**44.77**	
Open Source	Qwen3-VL-32B-Instruct	15.06	+78.31	13.74	+75.73	7.11	+28.67	11.93	+58.52	17.98	+25.44	12.14	+49.63
	AutoVSR(Qwen3-VL-32B-Instruct)	**93.37**		**89.47**		**35.78**		**70.45**		**43.42**		**61.77**	
	Llama-4-17B-128E	10.84	+42.77	10.23	+45.33	6.47	+18.53	7.39	+28.41	13.16	+22.37	9.16	+30.01
	AutoVSR(Llama-4-17B-128E)	**53.61**		**55.56**		**25.00**		**35.80**		**35.53**		**39.17**	
	GLM-4.6V-Flash	3.61	+74.70	8.19	+65.20	6.25	+30.17	8.52	+34.09	10.09	+16.66	7.34	+42.51
	AutoVSR(GLM-4.6V-Flash)	**78.31**		**73.39**		**36.42**		**42.61**		**26.75**		**49.85**	
Avg.	Base Model	14.89	+66.43	15.29	+64.49	10.68	+29.44	11.77	+42.86	11.84	+29.89	12.70	+44.37
	AutoVSR	**81.32**		**79.78**		**40.12**		**54.63**		**41.73**		**57.07**	

Improvement range across models: +30.01 to +59.45 percentage points.

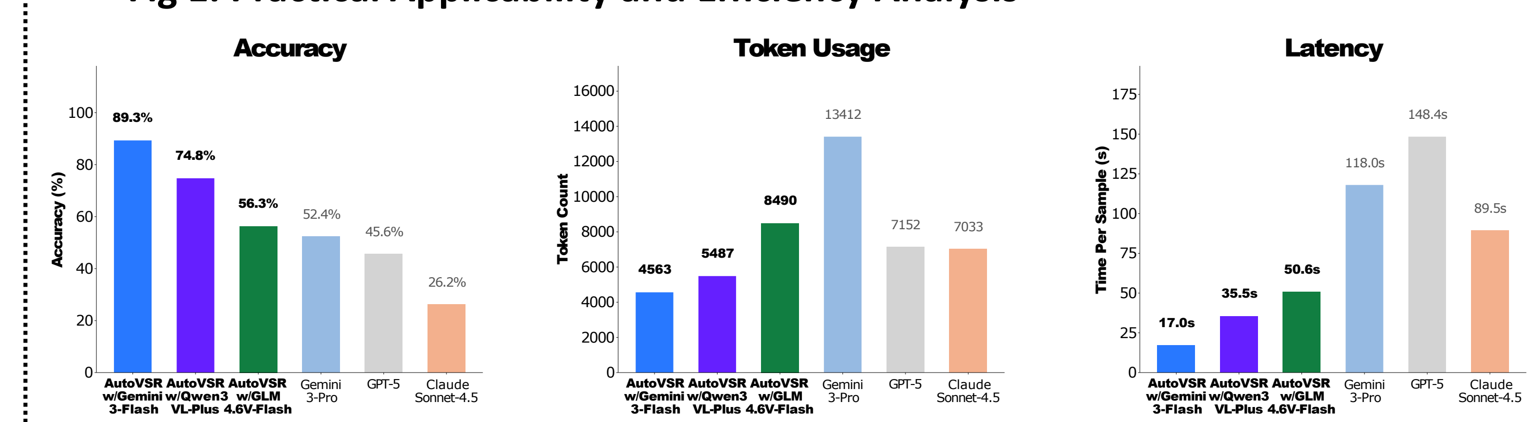
Benchmark: CircuitSense symbolic-expression-generation tasks, including transfer function generation and transient response expression generation across five circuit types.

Table 2. Performance Comparison between AutoVSR and CircuitSense.

Model	CircuitSense	AutoVSR	Improvement(%)
Transfer Function Generation			
Gemini-3-Flash	40.89	**82.85**	+ 41.96
Qwen3-VL-Plus	16.18	68.02	+ 51.84
GLM-4.6V-Flash	7.80	49.85	+ 42.05
Transient Response Generation			
Gemini-3-Flash	3.10	**85.84**	+ 82.74
GLM-4.6V-Flash	1.02	63.25	+ 62.23

Deterministic symbolic tools and executable intermediate representations yield much stronger performance than circuit-specific end-to-end reasoning.

Fig 1. Practical Applicability and Efficiency Analysis



AutoVSR with lightweight VLM backbones achieves higher accuracy with fewer tokens and lower inference runtime than expensive frontier VLMs.